

Instructions:

- **Due May 4, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as `ENME403_S26_HW10_yourname.pdf`.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning Objective: This homework develops students' ability to design and analyze state-space control systems using full-state feedback and observer-based methods. Students will assess controllability and observability, design state-feedback gains using pole placement, and evaluate how pole locations affect closed-loop performance and control effort. The assignment also introduces integrator augmentation for reference tracking and observer design for state estimation. Students will use MATLAB and Simulink to implement and validate controller and observer performance.

WARNING!!! Matlab assumes that the state-feedback control law is $u = -Kx$ and thus $\tilde{A} = A - BK$. However, we assume $u = Kx$ and thus $\tilde{A} = A + BK$. Therefore, you need to multiply the gains computed by Matlab functions by -1 .

Problem 1. Full state feedback (FSF) stabilizing controller. Consider the system

$$G(s) = \frac{s^2 + s - 6}{(s + 0.1)(s - 1.5)(s^2 + 2)}. \quad (1)$$

1. Using the Matlab function `tf2ss`, write $G(s)$ in the state-space form.
2. Determine whether (A, B) is controllable using the Matlab function `ctrb`.
3. Use `place` to compute the state feedback gain matrix K such that the closed-loop eigenvalues, that is, the eigenvalues of $A + BK$, are at $-3, -3.5, -3.75$, and -4 .
4. Simulate the closed-loop system in Simulink. Assume that $x(0) = [1 \ 2 \ 3 \ 4]^T$. Show all states and the input.
5. Repeat 3 to place the closed-loop poles at $-9, -9.5, -9.75$, and -10 . Repeat 4.
6. Explain how the control effort and the state response changes as the desired closed-loop pole locations change.

Problem 2. Observer. Consider the system described by (1).

1. Determine whether (A, C) is observable using the Matlab function `obsv`.
2. Design an observer with eigenvalues at $-9, -10, -11$, and -12 . Note that the observer is

$$\dot{\hat{x}} = A\hat{x} + Bu + F(y - C\hat{x}), \quad (2)$$

where $F = -(\text{place}(A^T, -C^T, p))^T$ and the vector p contains the desired observer eigenvalues.

3. Simulate the observer in Simulink. Assume that $x(0) = [1 \ 2 \ 3 \ 4]^T$ and, for all $t \geq 0$, $u(t) = 1 + \sin(0.5t)$. Plot the system and the observer states in the same figure.
4. Repeat Problem 3.3 with observer eigenvalues scaled by α . Set α to 0.1, 0.5, 2, and 10. How is the convergence of the observer state to the system state affected by increasing α ? What should be the optimum value of α ?

Problem 3. Output feedback stabilizing controller. Consider the system described by (1). The objective is to design an output feedback controller to stabilize the system.

1. Consider the full-state feedback stabilizing controller and the observer you designed in the previous problems. Note that the solution of Problem 1 gives the state feedback gain matrix K , and the solution of Problem 2 gives the observer gain matrix F . Compute the eigenvalues of $A + BK$ and $A - FC$.
2. Determine the state-space matrices for the output feedback controller computed by combining the full-state feedback controller and the observer. Recall that the output feedback controller is

$$\dot{\hat{x}} = (A + BK - FC)\hat{x} + Fy, \quad (3)$$

$$u = K\hat{x}. \quad (4)$$

3. Compute the controller transfer function $G_c(s)$. Note that the input to the controller is y and its output is u , therefore, $G_c(s) = G_{uy}(s)$.
4. Simulate the closed-loop system in Simulink. Plot all states and the control input, and comment on the system response.